

THE AI TOOLBOX · CLASS 02

What Is Research? Finding an AI Question

Turn "I'm interested in X" into a question you could actually study.

Merit Point Academy · 60 min · press **S** notes · → next ·  to comment

RECAP · WHERE WE LEFT OFF

Last class in 60 seconds

AI $\supset$ ML $\supset$ DL

Nested circles, not synonyms. Deep Blue was AI but not ML; ChatGPT is all three.

AI is everywhere

Face unlock, feeds, maps — each one a **model trained on data**, not magic.

AI is an instrument

A microscope for research. You own the **question** and the **judgement**.

Homework share: your AI application map — 2–3 volunteers, 30 seconds each: your domain, your favorite system, its **input** → **output**. Keep your map handy; today you'll turn it into a research question.

Where we're going

- **0–8'** Warm-up + the race no car could finish
- **8–20'** What research really is · what makes a question "researchable"
- **20–30'** Good vs. bad questions · the input → output → metric recipe
- **30–38'** Three ways to find a question: funnel · SCAMPER · literature
- **38–46'** Live demos: ChatGPT · Google Scholar · ScholarMind
- **46–54'** Two hands-on Python labs (with an AI helper)
- **54–60'** Your 3 questions, homework & wrap-up
- Quizzes &  comments throughout

By the end: say what makes a question researchable, tell a strong question from a weak one, and turn your own interest into a specific, measurable AI question.

WARM-UP · ACTIVATE

Quick check · tap an answer

Your phone's autocorrect keeps 'fixing' a slang word you like. You wonder why, read how autocorrect learns, then type the word ten times to see if it adapts. What did you just do?

Wasted ten minutes of your life

A complete micro-research project — observe, question, explore, test

Programming — you changed the phone's code

Nothing special; research needs a lab and a white coat

THE STORY

The race no car could finish

In 2004, DARPA offered **one million dollars** to any team whose car could drive itself 150 miles across the Mojave Desert. **Not one car finished.** The best managed 7.4 miles before getting stuck. People called it a failure. One year later — with the challenge sharpened and the teams focused on **specific, measurable sub-problems** — five cars finished the entire course. That second race launched the modern self-driving industry.



Speak this story

Neural voice · click to play / pause

2004's question was a wish: "can a car drive itself?" 2005's teams asked answerable questions — "can we keep GPS lock in a tunnel?", "can we tell a rock from a shadow at 30 mph?" — and the wish became engineering.

FROM CURIOSITY TO RESEARCH

Curiosity, made systematic

Every real study starts the way your autocorrect experiment did. Watch one curiosity grow up:

1

Observe

Parrots and mynas can mimic human speech. Odd!

2

Question

Why? How do they control their vocal sounds?

3

Explore

Read what's known: songbird vocal organs, brain regions.

4

Focus

"Is myna song learning linked to neural plasticity?"

Core insight: research is not a distant academic ritual — it is **innate curiosity transformed into purposeful, organized, systematic exploration**. The magic step is 4: narrowing wonder into one testable question.

CONCEPT 1

What research actually is

Not just Googling

Searching finds *existing* answers. Research **critically filters, evaluates, and integrates** — it doesn't passively receive.

A systematic loop

Clear question, scientific method, **reproducible** validation, honest analysis — a closed loop, not random trial-and-error.

Creates new knowledge

The goal: knowledge or insight that **didn't exist before** and can withstand review by others.

1

Ask the question

from observation & curiosity

2

Form a hypothesis

a testable expectation

3

Design the experiment

variables, samples, method

4

Evidence & analysis

collect data, conclude — or loop back

CHECK · WHAT RESEARCH IS

Quick check · tap an answer

Four classmates describe their weekend. Who actually did research?

Copied three websites about volcanoes into one document

Googled 'best phone 2026' and read the first answer

Wondered if music helps homework, formed a guess, and timed themselves for a week with and without music

Watched nine documentaries about space back-to-back

CONCEPT 2

What makes a question "researchable"

A research question is a **single, answerable question that a study can settle with evidence.**

- **Specific** — names a scenario, not a sweeping theme
- **Measurable** — you can collect data on it
- **Answerable** — within your time, skills, and resources



The picture test:

close your eyes and picture the **table of data** that would answer your question. Can't see it? Not a research question yet.

KEY IDEA — If you can't picture the data that would answer it, it's not a research question yet.

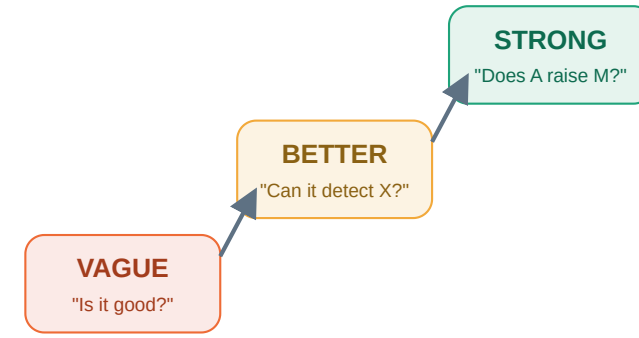
CONCEPT 3 · THE IDEA IN ONE PICTURE

The question ladder: vague → better → strong

- **Vague** — "Is self-driving good?" Nothing here can be measured.
- **Better** — "Can a model detect pedestrians at night?" Testable, but broad.
- **Strong** — "Do low-light training images raise night recall *without hurting day accuracy*?"

Bad questions are vague, unfalsifiable, or already answered. Good ones are narrow enough to test, open enough to be interesting.

each rung adds: a scenario, a measurement, a trade-off



Takeaway: a strong question names **what you'll measure** and **what would count as success**. Notice the strong rung even guards against a side effect — "without hurting day accuracy."

The AI-question recipe: input → output → metric

Abstracting turns a feeling into a model. Name three things and any interest becomes an AI question:



Input

What the model **sees** — an image, a sentence, a row of sensor data.

Night camera frames.



Output

What it **predicts** — a label, a number, a box.
"Pedestrian here" boxes.



Metric

How you'll **know it worked** — one honest number.

Recall: share of real pedestrians found.

KEY IDEA — **input + output + metric = an AI research question.** This triple is the seed of every ML study you'll run in this course — and it's exactly what AI research questions demand: data-centric, model-building, quantifiable.

Quick check · tap an answer

“Do low-light training images raise night-time pedestrian recall without hurting daytime accuracy?” — what makes this a **STRONG research question?**

It's about self-driving cars, which are exciting

It names the data, the measurement, and what success looks like — including a trade-off guard

It's longer than the other questions

It's impossible to answer, which makes it ambitious

"I want to make cars safer" — sharpened twice

A valuable vision, but too broad to act on. Real AV teams split it into layer-specific, measurable questions:

Perception layer

Q: How can a CNN detect pedestrians more accurately in **low-light night conditions**?

Input: night camera frames · **Output:** pedestrian boxes · **Metric:** accuracy & recall.

Prediction layer

Q: How can an RNN predict other vehicles' trajectories **5 seconds ahead** more accurately?

Input: recent motion tracks · **Output:** future path · **Metric:** error vs. the real trajectory.

Same recipe both times: **specific scenario + model + metric**. Abstract safety became two testable studies — this is our running AV example wearing its research-question hat.

"Use AI to help doctors" — sharpened twice

Lung nodule screening

Q: Can a deep model tell malignant from benign nodules in **low-dose CT** scans?

Metric: classification accuracy — while **controlling false positives**, so doctors aren't flooded with false alarms.

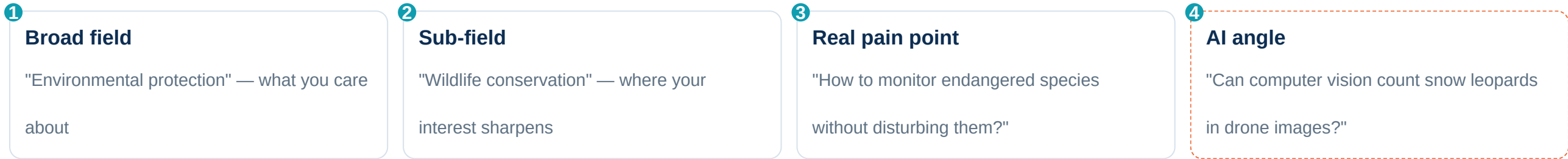
Brain tumor segmentation

Q: Can a U-Net segment MRI tumor regions **pixel-precisely** for surgery planning?

Metric: Dice coefficient — overlap between the model's region and expert annotation.

New field, same recipe. Notice each question carries its own **honest metric** — and a guard (false positives) just like the ladder's strong rung. Every domain you care about has questions shaped exactly like these.

The funnel: interest → sub-field → pain point → AI question



Result ▶ How can a YOLOv8 model detect and count snow leopards in complex mountainous drone images — despite vegetation, lighting changes, and camouflage?

Each step **narrows**: field → direction → problem → testable question. If your question still feels foggy, you stopped narrowing too early.

SCAMPER: seven angles on any process

Stuck? Point these seven prompts at any existing product or process — each angle can expose a research question.

S · Substitute

Can AI replace a manual process or tool?

C · Combine

Can AI + another technology create something new?

A · Adapt

Can a model from one field work in a totally different one?

M · Modify

Can we tune a model's structure or inputs to do better?

P · Put to other uses

What *else* could this existing AI system do?

E · Eliminate

Can AI remove a step, a cost, or a risk entirely?

R · Reverse — flip the roles or the order: instead of "librarians label books for readers," what if **readers' AI tags label books for the library**? The seventh angle is often the weirdest — and the most original.

Seven research questions hiding in a library



S

Can a robot **classify and reshelve** books as well as a librarian?



C

Face recognition + borrowing system = **"face-scan checkout"**?



A

Can e-commerce **"you may also like"** tech recommend library books?



M

Can search understand **"that dragon book from last summer"** — not just keywords?



P

Can borrowing data **predict community reading trends** and guide purchasing?



E

Can AI sorting **eliminate misplaced books** at the source?

One quiet building, seven possible studies — and each still needs the recipe: **what's the input, the output, the metric?** (Try it: the recommender's would be borrow history → suggested titles → % of suggestions borrowed.)

Quick check · tap an answer

Taking the 'you may also like' recommendation engine from online shopping and making it suggest library books is which SCAMPER move?

S — Substitute: AI replaces the librarian

C — Combine: merging two technologies

A — Adapt: borrowing a technique from a different field

E — Eliminate: removing a step

Stand on shoulders: search the literature

Why read papers?

Status quo — what's been done. **Gaps** — what hasn't. **Direction** — where the field says "future work should..."

The tools

Google Scholar — search everything.
Semantic Scholar & Connected Papers — see citation maps, find related work visually.

The 3-step loop

Anchor — find one good survey paper. **Extend** — follow its citation graph. **Lock** — read "Future Work" sections for unexplored directions.

The secret: the "Future Work" paragraph at the end of a paper is a list of research questions the authors **couldn't get to** — free, expert-approved gaps waiting for someone. Research gaps live at the boundary between known and unknown.

Google Scholar

Semantic Scholar

Connected Papers

LIVE DEMO 1

ChatGPT — brainstorm & refine

Not an answer machine today — a thinking partner. Watch it widen options, then sharpen the best one.

You ▶ I'm interested in AI in education. Brainstorm 5 research questions a high-schooler could actually test.

You ▶ Break 'AI essay grading' into 3 specific, researchable sub-questions.

You ▶ Rewrite my best question in input → output → metric form.

You ▶ What data would I need for this question – and where could I realistically get it?



chatgpt.com

LIVE DEMO 2

Google Scholar — has it been answered?

The researcher's search engine. Every result is a paper — use it to check your question against what's already known.

You ▶ Search: phishing email detection deep learning – skim the first page of titles.

You ▶ Open one paper. Read ONLY the abstract: what question did they ask? What did they find?

You ▶ Find the 'Future Work' or 'Limitations' paragraph – copy one unexplored direction.

You ▶ Now search your own question's keywords. Answered already? Great – narrow it or add a twist.



scholar.google.com

LIVE DEMO 3

ScholarMind AI — question + real datasets

Our research platform, step 1 of the workflow from Class 1: it grounds your topic in real, downloadable datasets and drafts the brief.

You ▶ Topic: 'wildlife + drones' → propose 3 testable research questions.

You ▶ It searches real datasets (Hugging Face, UCI, Kaggle) – check: does data for YOUR question actually exist?

You ▶ Analyze the best question: what data, model, and metric does it suggest?

You ▶ Generate the research brief – this brief.md is what a coding agent builds from in later classes.



academic.hopeembark.org

Build a research-question quality checker

Can code judge a question? A little. This checker looks for the **ingredients of a strong question** — data words, metric words, vague words. Run it on the ladder, then on **your own** question.

```
# A tiny research-question checker.
# Strong questions name their DATA and their METRIC - and avoid mush.

DATA_WORDS = ["image", "images", "photo", "audio", "text", "dataset",
              "video", "recordings", "scans", "sensor", "data"]
METRIC_WORDS = ["accuracy", "recall", "precision", "error", "f1",
               "dice", "rate", "percent", "score"]
TASK_WORDS = ["detect", "predict", "classify", "segment",
              "recognize", "count", "recommend", "translate"]
VAGUE_WORDS = ["good", "bad", "smart", "smarter", "best ever",
               "replace humans", "change the world"]

def check(question):
    q = question_lower()
```



Run



Explain



Debug



Improve

Python loads on first Run

► Run the code to see the output here.

AI Assistant

Haiku · fastest

Hi! Edit the code and hit **Run**. Stuck? I can **Explain**, **Debug**, or **Improve** it — or just ask me below.

Ask about this code...

Ask

Actually answer the strong question (toy data)

The strong question promised an answer from evidence — so let's get one. Two toy models, ten test photos each: did low-light training **raise night recall** without **hurting day accuracy**?

```
# The STRONG question, answered with (toy) data:  
# "Do low-light training images raise night recall  
# without hurting day accuracy?"  
# Each result: (time_of_day, found_the_pedestrian?) 1 = found, 0 = missed
```

```
model_A = [ # trained on daytime photos only  
    ("day", 1), ("day", 1), ("day", 1), ("day", 1), ("day", 0),  
    ("night", 1), ("night", 0), ("night", 0), ("night", 0), ("night", 0),  
]  
model_B = [ # same model + extra low-light training photos  
    ("day", 1), ("day", 1), ("day", 1), ("day", 0), ("day", 1),  
    ("night", 1), ("night", 1), ("night", 1), ("night", 0), ("night", 1),  
]
```



Run



Explain



Debug



Improve

Python loads on first Run

▶ Run the code to see the output here.

AI Assistant



Haiku · fastest



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Ask

YOUR PROJECT

Write three research questions

Take a topic you love — ideally from your Class-1 application map — and write **three questions at rising quality**: one simple, one medium, one strong.

- Pick your topic (any domain — sport, music, climate, games...)
- Write a **simple**, a **medium**, and a **strong** AI question
- Label *why* each sits at its level — which ingredient is missing or present?

Deliverable

3 labeled research questions, each with a one-line justification.

Success looks like

The strong one names an **input**, an **output**, and a **metric** — and is answerable with data you could realistically get.

BUILD IT NOW · HANDS-ON

Draft, sharpen, and reality-check your questions

1. Pick your topic — open your Class-1 application map for candidates
2. SCAMPER it, or run the funnel: field → sub-field → pain point → AI angle
3. Ask ChatGPT: "Give me 5 research questions about <topic>" — then sort them weak → strong
4. Rewrite the best one in **input** → **output** → **metric** form
5. Check Google Scholar — has someone already answered it? Narrow or twist.

ChatGPT

Google Scholar

ScholarMind

Tip: if Scholar shows it's been done, that's **good news** — the topic matters and data exists. Add a twist: a new scenario, a new input, a stricter guard.

What a finished strong question looks like

✗ Not research questions

"How can we make AI smarter?" — no domain, no data, no metric.

"Will AI replace humans?" — a debate topic; no experiment can settle it.

✓ A real one (student-reachable)

"Phishing emails keep mutating past old filters. Can an **attention-based** model beat a classic keyword filter on **detection accuracy and F1** — especially on **never-seen-before** phishing styles?"

Specific domain

Phishing detection — not "cybersecurity" in general.

Honest metrics

Accuracy, F1 — and a hard test: unseen phishing styles.

Real value

A live security pain point — the answer would matter.

Today's vocabulary

Research question — a single, answerable question a study can settle with evidence.

Metric — the honest number that says how well it worked (accuracy, recall, error...).

Literature review — systematically reading prior work to map the known and unknown.

Hypothesis — your testable expectation of the answer, made before looking.

Recall — of all the real positives out there, the share your model actually found.

Research gap — a question at the edge of the known that nobody has answered yet.

RECOMMENDED HOMEWORK

Before next class

1. **Finalize your three questions** in a short doc — next class opens with group peer review. Make the strong one pass the recipe: input, output, metric.
2. **Read 3 abstracts** on Google Scholar near your strong question. For each, jot one line: their method, their finding, and one gap you noticed.
3. **Re-run Lab 1** on your final question — did it score STRONG? Should it?

★ Stretch (optional)

Skim Kaggle's "**Google QUEST Q&A Labeling**" — watch how a mushy question ("is this a good answer?") was turned into **measurable labels**. That's today's lesson, industrial-scale.

[kaggle.com](https://www.kaggle.com/google/quest) · Google QUEST

Submit the doc in Google Classroom (Classwork → Assignments). Stuck? Post your draft question on any slide with  — your instructor will see it.

TODAY'S LINE TO REMEMBER

"The formulation of a problem is often more essential than its solution."

— Albert Einstein

Ask first. Answers are downstream of good questions.



Speak this

WRAP-UP

Three things to keep

- Research = curiosity made **systematic** — a loop, not a lucky search
- A strong question names its **input, output, and metric** — and guards a trade-off
- Generate widely (funnel · SCAMPER · ChatGPT), then **verify ruthlessly** (Scholar · data check)

Exit ticket: post your strongest question in  on this slide — first feedback is free.

Next · Class 3 — the AI research pipeline: the road your question now travels.

BACKUP MATERIAL

Extra slides — if there's time

Deeper dives and extra practice. Skip in a tight hour; pull up on demand or for a longer session.

The value of a question

three dimensions of worth

Question first-aid clinic

fix the four common failures

Bonus quiz

one more check

What makes a question worth asking?



Academic value

Does it fill a theoretical gap or push the boundary of what's known?



Practical value

Does it address a real social need or a genuine pain point someone has today?



Innovation value

Does it bring a new method, a new angle, or a new framing to an old problem?

Self-check: score your strong question 1–5 on each dimension. It doesn't need three 5s — but three 1s means keep looking. Most great student projects score high on **practical** value first.

The four ways questions fail — and the fix

1 · Too broad

"How can AI improve education?" → **Fix:** run the funnel one more turn.
Which subject? Which task? Which learners?

2 · Not measurable

"Can AI make music more beautiful?" → **Fix:** find the proxy metric. Listener ratings? Skip rate? Define the number.

3 · Already answered

Scholar's first page settles it → **Fix:** add a twist — new scenario, new data, stricter guard. ("...but does it hold for *Mandarin* emails?")

4 · Data out of reach

"...using hospital MRI records" you'll never access → **Fix:** hunt public datasets first (Kaggle, UCI, Hugging Face — or ScholarMind), then shape the question around what exists.

Quick check · tap an answer

“Will AI replace humans?” is a weak research question mainly because...

It isn't about technology

Nobody is interested in the answer

No experiment or dataset could settle it — it isn't measurable or falsifiable

It's too short to be a question